North American Bat Monitoring Program in Mount Revelstoke and Glacier National Parks

Manual Verification Data Summary 2025

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1 Executive Summary

2 Land Acknowledgement

Biodiversity Pathways respectfully acknowledges that our work takes place on Treaty 8 and Douglas Treaties Territories as well as the traditional and unceded territories of First Nations and Métis Peoples across all regions of British Columbia, whose histories, languages, and cultures are deeply connected to the biodiversity we monitor. We acknowledge the traditional teachings of the lands that we work on, and that reciprocal, meaningful, and respectful relationships with Indigenous peoples make our work possible. We are deeply grateful for their stewardship of these lands, and we are committed to supporting Indigenous-led monitoring programs, while learning Indigenous ways of knowing, being, and doing.

3 Background

3.1 Overview of NABat and the NNW Bat Hub

The North American Bat Monitoring Program (NABat) is a large-scale coordinated effort to monitor bat species across North America using standardized protocols and a unified sample design (Loeb et al. 2015). NABat was established to address the gaps in knowledge and lack of long-term studies of bat species across Mexico, USA, and Canada. The program is administered by the US Geological Survey (USGS), coordinated by the Canadian Wildlife Health Cooperative (CWHC) in Canada, and implemented by the North by Northwest Bat (NNW) Hub in British Columbia, Alberta, and S.E. Alaska.

3.2 NABat Monitoring at Mount Revelstoke and Glacier National Parks

Mount Revelstoke and Glacier National Parks have monitored two NABat grid cells since 2017 and have collected 9 years of data. Both grid cells are in southeastern BC. Mount Revelstoke contains 2 stationary bat detectors and 1 transect route while Glacier contains 3 stationary detectors, all detectors were deployed in primarily forest habitat.

In 2025, the transect route was monitored using two acoustic recording units, the Anabat and the EchoMeter Touch Pro, to evaluate differences in species detection between the two detectors. This comparison will inform the park's transition to the EchoMeter Touch Pro for future data collection.

4 Methodology

Full-spectrum recordings from the sampling periods were processed using two automatic classifiers: Kaleidoscope's Bats of North America 5.4.0 classifier and Sonobat 3.0's Southeastern British Columbia classifier. Based on documented species ranges, manual identification efforts focused on 10 species: Townsend's big-eared bat (*Corynorhinus townsendii*), Big Brown Bats (*Eptesicus fuscus*), Silver-haired Bats (*Lasionycteris noctivagans*), Hoary Bat (*Lasiurus cinereus*), California Myotis

(*Myotis californicus*), Long-eared myotis (*Myotis evotis*), Little Brown Myotis (*Myotis lucifugus*), Northen Myotis (*Myotis spententrionalis*), Long-legged Myotis (*Myotis volans*), and Yuma Myotis (*Myotis yumanensis*).

The analysis workflow followed processing standards established by the North American Bat Monitoring Program (NABat) (Reichert et al. 2018). Only recordings that were identified as a bat by Kaleidoscope were eligeble to be manually verified. Species identifications were validated using reference call parameters described by Szewczak (2018), Slough et al. (2022), and Solick (2022), in accordance with NABat manual vetting protocols (Reichert et al. 2018). All species codes used in the analysis are summarized in Appendix A

5 Results and Discussion

5.1 Species detection

All previously detected species were identified again across sites. Feeding buzzes were recorded in the NW quadrant of the Mount Revelstoke grid cell and the SW quadrant of the Glacier grid cell Table 1, and one social call was recorded in the NW quadrant of the Mount Revelstoke cell.

Results from the full dataset, including automated ID, indicate that the most commonly recorded species across both grid cells is the Little Brown Bat (*Myotis lucifugus*; MYLU) (Figure 1, Figure 2).

Table 1: Manual verification summary results across grid cells and sites.

Site	COTO	EPFU	LACI	LANO	MYCA	MYEV	MYLU	MYSE	MYVO	MYYU	MYOTIS	40KMYO	HIF	LOF	FB	SC
241002																
NE	0	1	3	4	0	3	14	1	7	0	1	0	0	0	0	0
NW	5	20	7	29	8	26	19	3	17	1	3	2	1	4	11	1
Transects	0	8	10	24	1	11	1	0	6	0	0	0	0	1	4	0
271731																
SE1	1	0	0	6	5	26	6	1	27	0	1	3	0	0	0	0
SW	5	0	1	3	2	30	7	2	29	0	2	7	0	0	10	0
Total	**11**	**29**	**21**	**66**	**16**	**96**	**47**	**7**	**86**	**1**	**7**	**12**	**1**	**5**	**25**	**1**

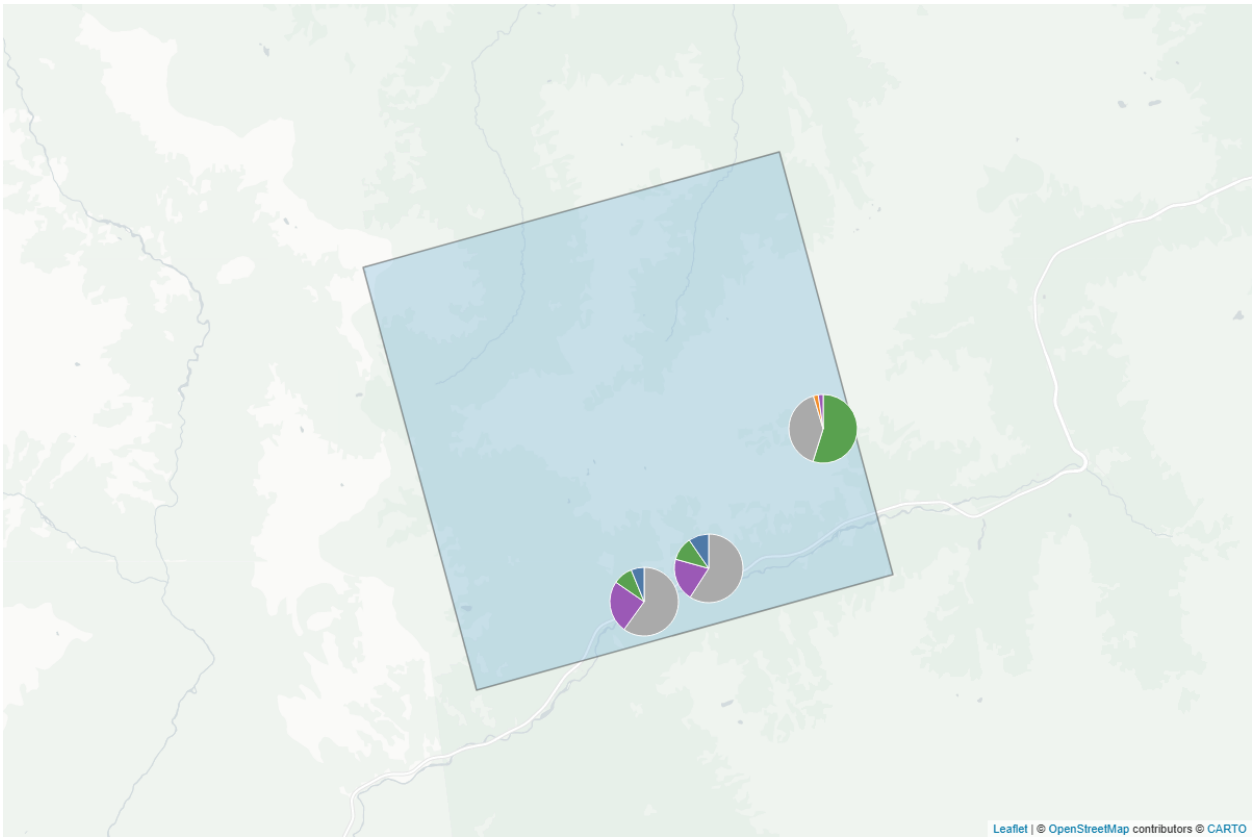

Figure 1: Sampled sites in Glacier National Park (GRTS 271731). Proportion of species bat passes at each site. Other represents tags that were identified as species couplets or species groups

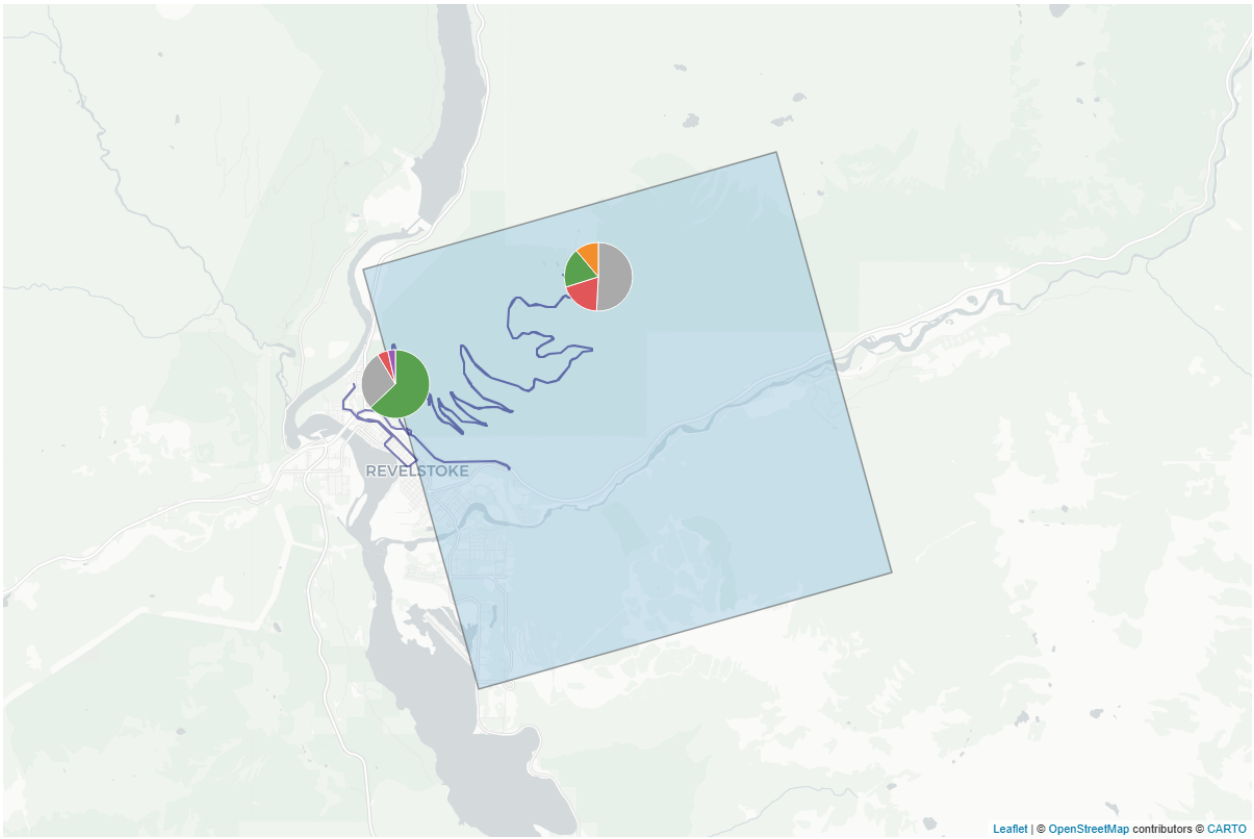

Figure 2: Sampled sites and driving transect route in Mount Revelstoke National Park (GRTS 241002). Proportion of species bat passes at each site. Other represents tags that were identified as species couplets or species groups

5.2 Paired deployments for the transect

The EchoMeter Touch Pro, which records in full spectrum, produced bat calls that were more readily identifiable to the species level compared to the Anabat zero-crossing recorder. However, this difference was not substantial between units, suggesting that transitioning to the EchoMeter Touch Pro is unlikely to require a correction factor. The province-wide analysis, which includes up to 20 sites and transects with paired deployments, will examine the use of correction factors more closely. Should that larger dataset suggest otherwise, a correction factor can be revisited when trend estimates are calculated for this grid cell.

5.3 Data Availability

All tag data has been uploaded to NABat under project name MRG-NABat Monitoring (ID: 16822). Recordings and tag data can also be found in Wildtrax under the project name Bat community - Long Terms Monitoring - NaBat - 2024/25

6 Appendix A

Code	Common Name	Scientific Name	Definition
Species			
COTO	Townsend's big-eared bat	*Corynorhinus townsendii*	
EPFU	big brown bat	*Eptesicus fuscus*	
LANO	silver-haired bat	*Lasionycteris noctivagans*	
LACI	hoary bat	*Lasiurus cinereus*	
MYCA	California myotis	*Myotis californicus*	
MYEV	long-eared myotis	*Myotis evotis*	
MYLU	little brown bat	*Myotis lucifugus*	
MYSE	northern long-eared bat	*Myotis septentrionalis*	
MYVO	long-legged myotis	*Myotis volans*	
MYYU	Yuma myotis	*Myotis yumanensis*	
Cou-plet/Group-ing			
EPFULANO	big brown bat, silver-haired bat	*Eptesicus fuscus, Lasionycteris noctivagans*	
EPFULABO	big brown bat, eastern red bat	*Eptesicus fuscus, Lasiurus borealis*	
EPFULACI	big bat, hoary bat	*Eptesicus fuscus, Lasiurus cinereus*	
EPFUMYLU	big brown bat, little brown bat	*Eptesicus fuscus, Myotis lucifugus*	
LABOMYLU	eastern red bat, little brown bat	*Lasiurus borealis, Myotis lucifugus*	
LACILANO	hoary bat, silver-haired bat	*Lasiurus cinereus, Lasionycteris noctivagans*	
MY-CAMYYU	California myotis, Yuma myotis	*Myotis californicus, Myotis yumanensis*	
MYEVMYSE	long-eared myotis, northern long-eared bat	*Myotis evotis, Myotis septentrionalis*	
MYLUMYSE	little brown bat, northern long-eared bat	*Myotis lucifugus, Myotis septentrionalis*	

Code	Common Name	Scientific Name	Definition
MY-LUMYVO	little brown bat, long-legged myotis	*Myotis lucifugus, Myotis volans*	
MY-LUMYYU	little brown bat, Yuma myotis	*Myotis lucifugus, Myotis yumanensis*	
Frequency Class			
Myotis			Myotis species
40kMyo			Various species of Myotis with pulses that have a minimum frequency of approximately 35-45 kHz.
HighF or HiF			Various species with pulses having a minimum frequency higher than ~30 kHz.
LowF or LoF			Various species with pulses having a minimum frequency lower than ~30 kHz.

Literature Cited

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